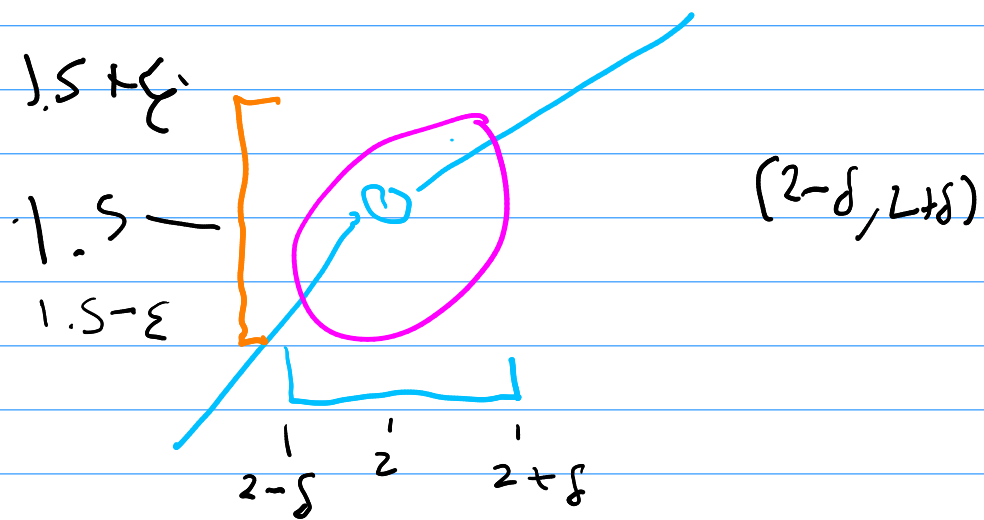
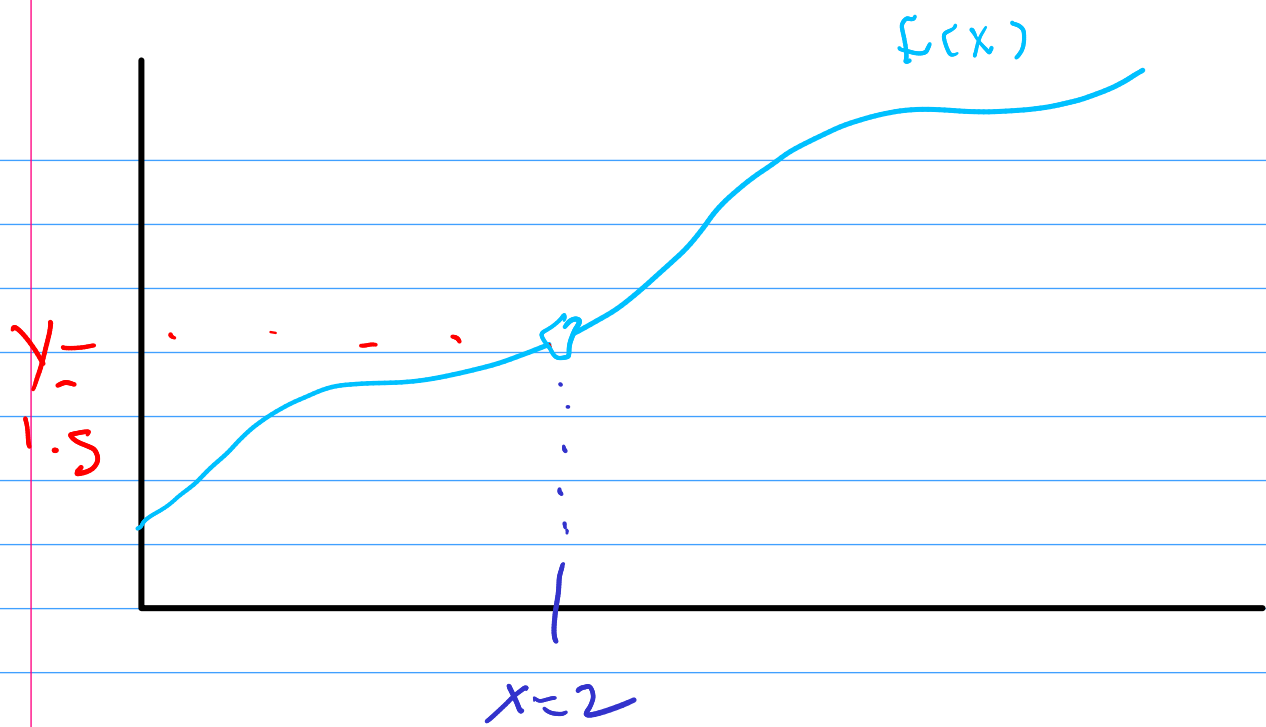




If we're close enough in x ,
 y stays within a certain small range

$$y = \frac{1}{2}(x+1)$$

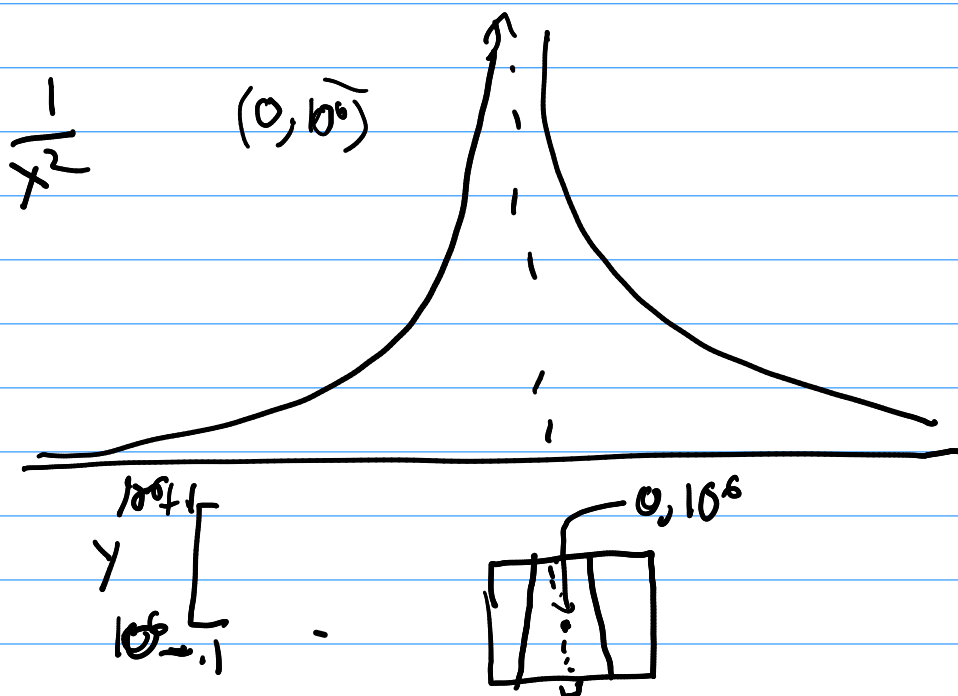
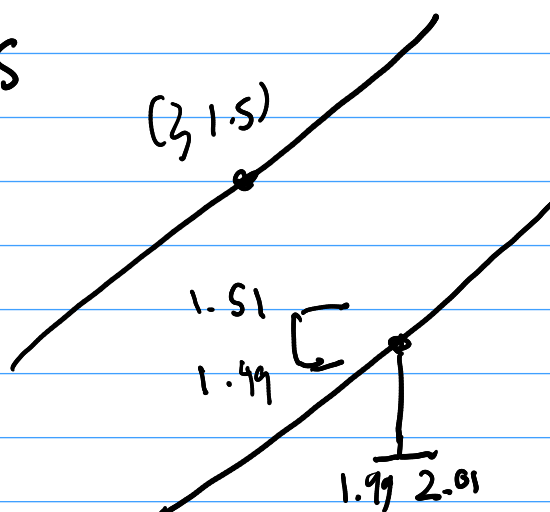
$$\lim_{x \rightarrow 2} (y) = 1.5$$

$$\frac{1}{2}(2.01+1)$$

$$1.505$$

$$\frac{1}{2}(1.99+1)$$

$$1.495$$



$$\lim_{x \rightarrow 1} \frac{(2x^2 - 2)}{2x^2 + x - 3} \quad \frac{2(1) - 2 = 0}{2(1) + 1 - 3 = 0}$$

$$\frac{2(x-1)(x+1)}{(x-1)(2x+3)} \quad \frac{2(2)}{5} = \frac{4}{5}$$

δ : for every x between $1-\delta$ to $1+\delta$,
 y is within 0.01 of $\frac{4}{5}$

try: $\delta = 0.01$

0.99 1.01

$$f(0.99) = 0.79919...$$

$$f(1.01) = 0.801...$$

